PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
X = [[25, 30, 600, 1], [45, 80, 750, 0], [35, 50, 680, 1], [28, 25, 550, 2],
    [50, 90, 780, 0], [23, 20, 520, 1], [40, 70, 720, 1], [32, 45, 640, 2],
    [55, 95, 800, 0], [30, 28, 580, 3], [38, 65, 700, 1], [27, 35, 610, 0],
    [48, 85, 760, 0], [26, 22, 540, 2], [42, 75, 730, 1], [33, 40, 630, 2],]
y = [0, 1, 1, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0]
features = ["age", "income", "credit_score", "existing_loans"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=1)
model = DecisionTreeClassifier(criterion="entropy", random_state=1)
model.fit(X_train, y_train)
print("Accuracy:", accuracy_score(y_test, model.predict(X_test)))
print("\nDecision Tree Rules:")
print(export_text(model, feature_names=features))
customer = [[36, 60, 710, 1]]
decision = model.predict(customer)[0]
print("New customer", customer[0], "->", "APPROVE" if decision == 1 else "REJECT")
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ================
Accuracy: 1.0

Decision Tree Rules:
|--- credit_score <= 665.00
|   |--- existing_loans <= 0.50
|   |   |--- class: 1
|   |--- existing_loans >  0.50
|   |   |--- class: 0
|--- credit_score >  665.00
|   |--- class: 1

New customer [36, 60, 710, 1] -> APPROVE
>>>
```

Ln: 17 Col: 0